ICE / ENGINEERING CALCULATION / 0.2.0

Gasoline internal combustion engine

Toyota 1ZR-FAE — geometric baseline
Auris 1.6 Valvematic / European 2015 range

English version · HTML / PDF · native engdoc

1,598 L	59,42 kW	129 N·m	6,647 MPa
Displacement	Calculated power	Calculated torque	Maximum pressure

Model operating point: n = 4 400 min⁻¹; α = 1; рвп = 98 kPa; θ_0 = 338°; Δθ = 55°. Timing and masses are assumptions.

> **Uncalibrated calculation model.** The Euro 6 designation refers to the manufacturer's vehicle range [1], not these simulation results. Adopted assumptions, numerical residuals and departures from specifications appear below.

Version 0.2.0 · 23.09.2026 · Full 720° cycle · Vector figures
Contents, formulas, tables and charts are generated from one project state.

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1. Purpose, prototype and limitations

Calculation report and interactive document for a four-stroke gasoline engine. This version is English. The first A1 sheet collects six main groups of calculated diagrams; it is not an engine assembly drawing.

Adopted model reference: Toyota Auris 1.6 Valvematic, European 2015 range, for which Toyota declares Euro 6. Reference rated points: 97 kW at 6400 min^{-1} and 160 N·m at 4400 min^{-1} [1]. The 1ZR-FAE geometry comes from the Auris 2012 technical description: 80,5 × 78,5 mm, ε = 10,7, four cylinders [2]. Applying geometry from the same family to the calculated 2015 variant is an explicitly adopted baseline, not confirmation of every internal detail of the Euro 6 version.

The calculation is single-zone, spatially averaged and uncalibrated. It determines mass, energy, pressure, temperature, heat transfer, work, fuel consumption and slider-crank loads. It does not calculate knock, atomisation, local mixture composition, NO_x, CO, HC, particulates, catalyst warm-up or a certification cycle. Modelling a Euro 6 vehicle does not demonstrate compliance of the calculation with Euro 6.

Every parameter is classified as manufacturer data, literature example, adopted assumption or project choice. Unknown Valvematic timing, valve lifts and component masses are not represented as factory specifications. Deviations from rated points appear without automatic fitting.

The thermal-calculation example supplies the sequence: working fluid → intake → compression → combustion → expansion and exhaust → indicated and brake performance [3]. A new numerical calculation replaces corrections to old figures. Semi-empirical heat-release curves require adaptation to the specific engine [4].

2. Geometric parameters

D is cylinder bore, m; S is piston stroke, m; ε is compression ratio; z is cylinder count. Values follow the geometric source [2]. L is connecting-rod length, m; it is an assumption, not a Toyota specification.

D — Cylinder bore: **0,0805 m**. Manufacturer data [2]

S — Piston stroke: **0,0785 m**. Manufacturer data [2]

ε — Geometric compression ratio: **10,7** . Manufacturer data [2]

z — Number of cylinders: **4** . Manufacturer data [2]

L — Connecting rod centre distance: **0,14 m**. Adopted assumption

n — Rotational speed, revolutions per second: **73,3333 s^{-1}**. Project selection

Area acted on by the pressure difference.

$$A_p = \pi \cdot D^2 / 4 \quad (2.1)$$

= 3,141593 · (0,0805 m)^2 / 4

= 0,00508958 m²

Volume swept by the piston during a stroke.

$$V_h = A_p \cdot S \quad (2.2)$$

= (0,00508958 m²) · (0,0785 m)

= 3,995·10^{-4} m³

Check of the geometric baseline's displacement.

$$V_d = z \cdot V_h \quad (2.3)$$

= (4) · (3,995·10^{-4} m³)

= 0,00159813 m³

Minimum-volume calculation; chamber surfaces in the heat-transfer model are assumed flat.

$$V_c = V_h / (\varepsilon - 1) \quad (2.4)$$

= (3,995·10^{-4} m³) / ((10,7) - 1)

= 4,119·10^{-5} m³

Centred slider-crank geometry.

$$r = S / 2 \quad (2.5)$$

= (0,0785 m) / 2

= 0,03925 m

This symbol is not used for the excess-air ratio.

$$\lambda_k = r / L \quad (2.6)$$

= (0,03925 m) / (0,14 m)

= 0,280357

n in the formula uses s^{-1}; input min^{-1} values are explicitly divided by 60.

$$\omega = 2 \cdot \pi \cdot n \tag{2.7}$$

$= 2 \cdot 3{,}141593 \cdot (73{,}3333\ s^{-1})$

= 460,767 rad/s

Characteristic speed for heat-transfer and mechanical-loss estimates.

$$c_m = 2 \cdot S \cdot n \tag{2.8}$$

$= 2 \cdot (0{,}0785\ m) \cdot (73{,}3333\ s^{-1})$

= 11,5133 m/s

3. Working fluid and fuel

The initial variant uses an effective gas with constant R = 287 J/(kg·K) and variable heat capacity. This is a model approximation, not a tabulated combustion-product composition. Air or diesel-example heat capacity is not directly adopted as the properties of actual combustion products.

α — Excess-air ratio: **1** . Project selection

l_0 — Stoichiometric air-to-fuel mass ratio: **14,956** . Literature example [3]

H_u — Lower heating value of the model gasoline: **4,400·10⁷ J/kg**. Literature example [3]

η_c — Fraction of chemical energy converted to heat: **0,98** . Adopted assumption

Literature fuel baseline: mass fractions C = 0,855; H = 0,145; O = 0; l_0 = [(8/3)C + 8H − O]/0,23 ≈ 14,956 kg air/kg fuel; H_u = 44 MJ/kg [3]. This represents gasoline without oxygenates. Actual E5/E10 composition and heating value require separate inputs; the Euro 6 label does not specify ethanol content.

Adopted effective relation: $c_v(T)$ = 675 + 0,25T, J/(kg·K); u(T) = 675T + 0,125T², J/kg; h(T) = u(T) + RT; k(T) = 1 + R/c_v(T). (3.1) Coefficients are an explicit first-level assumption. The model excludes dissociation, changes in molar mass during reaction and chemical equilibrium.

Fuel mass is already included in the fresh homogeneous mixture. Combustion adds energy to the gas, not a second copy of fuel mass. η_c specifies incomplete energy conversion; changing α does not automatically model pollutant concentrations.

The initial version does not support rich mixtures (α < 1): without a CO/H_2 and incomplete-chemistry model, oxygen-deficient fuel cannot be assigned almost complete heat release. The interface restricts α to 1,0–1,1.

4. Gas exchange and filling

Angle θ = 0° is gas-exchange TDC; 180° is intake BDC; 360° is combustion TDC; 540° is expansion BDC; 720° is the next gas-exchange TDC. Intake starts at 0°, exhaust closes at 720°. The baseline has no overlap.

Valve lift is h_v(θ) = h_max·sin²[π(θ−θ_open)/(θ_close−θ_open)] inside the open interval and zero outside it. A_v = 2·min(πd_vh_v, πd_v²/4). (4.1) Two valves of each type per cylinder are assumed. This is not measured Valvematic kinematics; lift may be changed manually.

The pressure difference determines flow direction for each valve. Flow is choked at p_d/p_u ≤ [2/(k+1)]^{k/(k−1)}. Above the critical ratio, the isentropic flow formula is used; below it, its value at M = 1 [5].

ṁ = C_d A_v p_u/√(RT_u) · Ψ; Ψ = √{2k/(k−1)·[(p_d/p_u)^{2/k}−(p_d/p_u)^{(k+1)/k}]} for unchoked flow; Ψ = √k·[2/(k+1)]^{(k+1)/[2(k−1)]} for choked flow. (4.2) p_u and T_u come from the upstream side; reverse flow is not clipped.

ṁ_fresh = ṁ_in,fresh − Y_fresh·ṁ_out; Y_fresh = m_fresh/m. (4.3) At the start of a repeated cycle, residual mass has zero freshness marker. Fresh-mixture mass is fixed at intake closure to determine fuel per cycle. This marks mass origin, not a chemical component.

m_a — Gas mass at intake closure: **4,193·10⁻⁴ kg**. Numerical snapshot

x_r — Residual mass fraction at intake closure: **0,0437497** . Numerical snapshot

Separating fresh charge from residual gas.

$$m_{fresh} = m_a \cdot (1 - x_r) \quad (4.4)$$

= (4,193·10⁻⁴ kg) · (1 - (0,0437497))

= 4,009·10⁻⁴ kg

Including fuel in the fresh-mixture mass.

$$m_f = m_{fresh} / (1 + \alpha \cdot l_0) \quad (4.5)$$

= (4,009·10⁻⁴ kg) / (1 + (1) · (14,956))

= 2,513·10⁻⁵ kg

T_a — Temperature at intake closure: **365,071 K**. Numerical snapshot

p_a — Pressure at intake closure: **108 294 Pa**. Numerical snapshot

Quantity at intake closure	Value
p_a, MPa	0,1083
T_a, K	365,1
m_a, mg	419,3
m_f, mg/cycle	25,13
Residual mass fraction, %	4,375

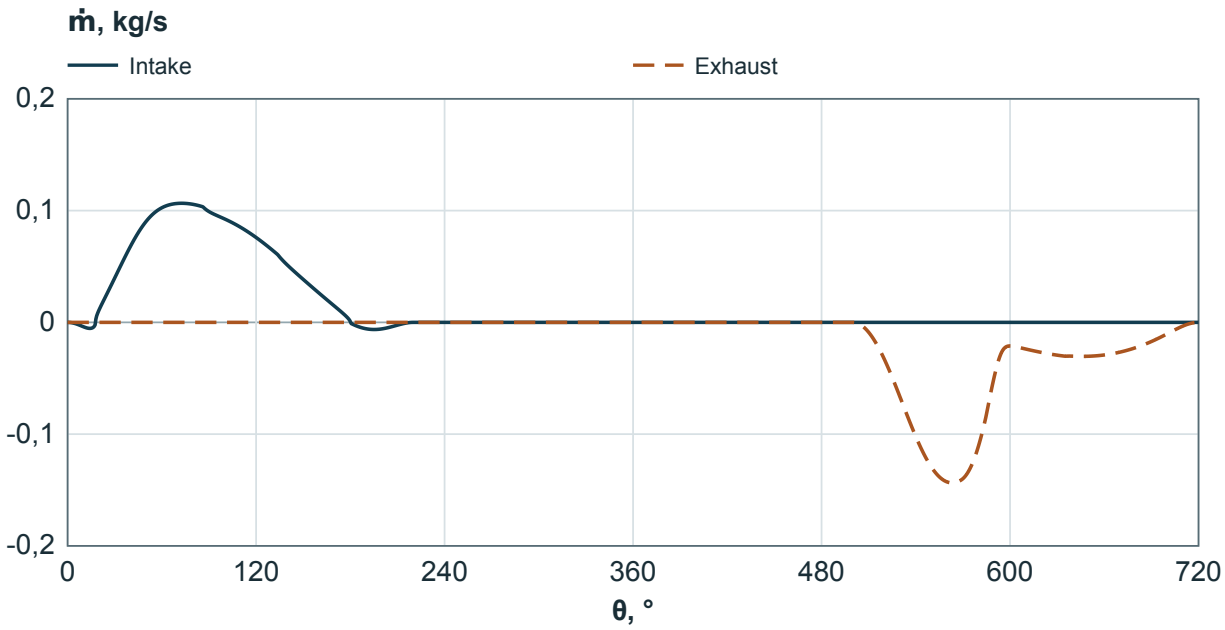


Figure 4.1 — Valve flow; positive sign means into the cylinder

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ_0 = 338°; Δθ = 55°. Timing and masses are assumptions.

5. Compression and energy balance

Mass and energy are integrated from intake opening to exhaust closure. Between intake closure and exhaust opening, the cylinder is closed to mass transfer. Ring blow-by is excluded from the initial variant.

dm/dθ = (Σṁ_in − Σṁ_out)/(6n_rpm). (5.1) Angle is measured in degrees; n_rpm is numerical speed in min⁻¹. When radians are used, the divisor is ω; these scales are not mixed.

dU/dθ = dQ_c/dθ − dQ_w/dθ − p·dV/dθ + [Σṁ_in h_in − Σṁ_out h]/(6n_rpm). (5.2) U = m u(T); p = mRT/V. Q_w is positive from gas to wall. Negative heat flow during cold intake is permitted.

During compression, Q_c = 0. A polytropic exponent is not assigned to force a desired p_c; pressure follows the energy equation, kinematics and heat transfer. The pre-combustion state appears below; it differs from the TDC state after some heat has been released.

p_c — Pressure immediately before heat release: **1,443·10⁶ Pa**. Numerical snapshot

T_c — Temperature immediately before heat release: **715,845 K**. Numerical snapshot

Explicit fourth-order Runge–Kutta integration is used. The grid includes intake closure, start/end of combustion and exhaust opening independently of the main step. Periodic state is sought by repeating cycles; the criterion checks mass and energy at the start and end of 720°. Step checks at 1° / 0,5° / 0,25° appear in the protocol rather than an unsupported accuracy declaration.

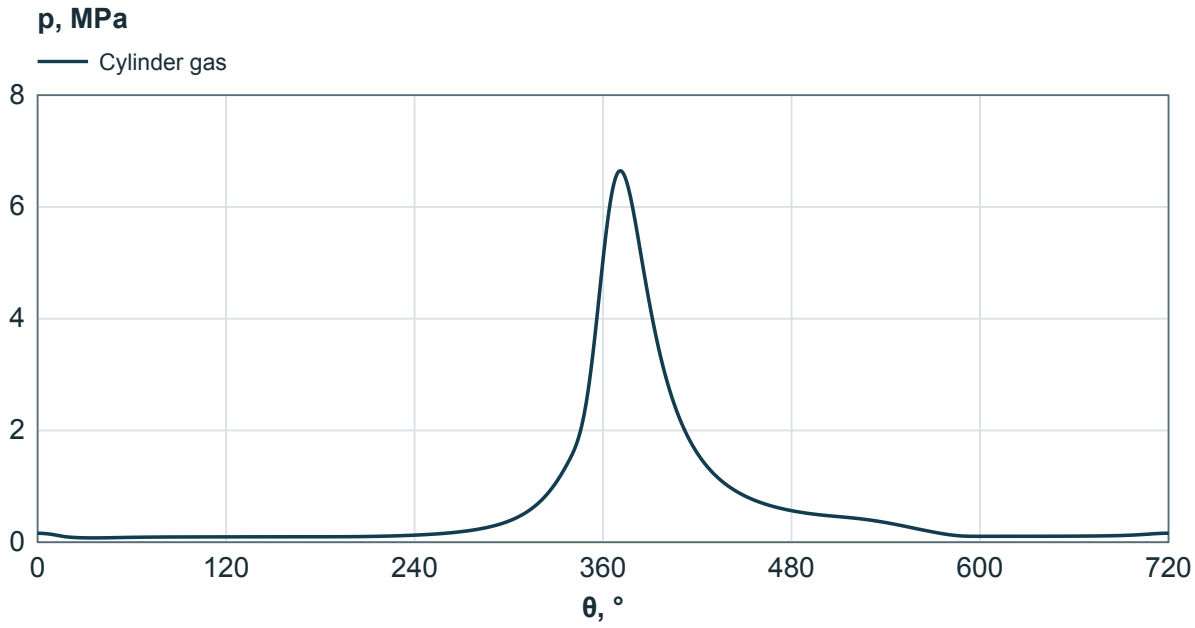


Figure 5.1 — Pressure versus crank angle

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

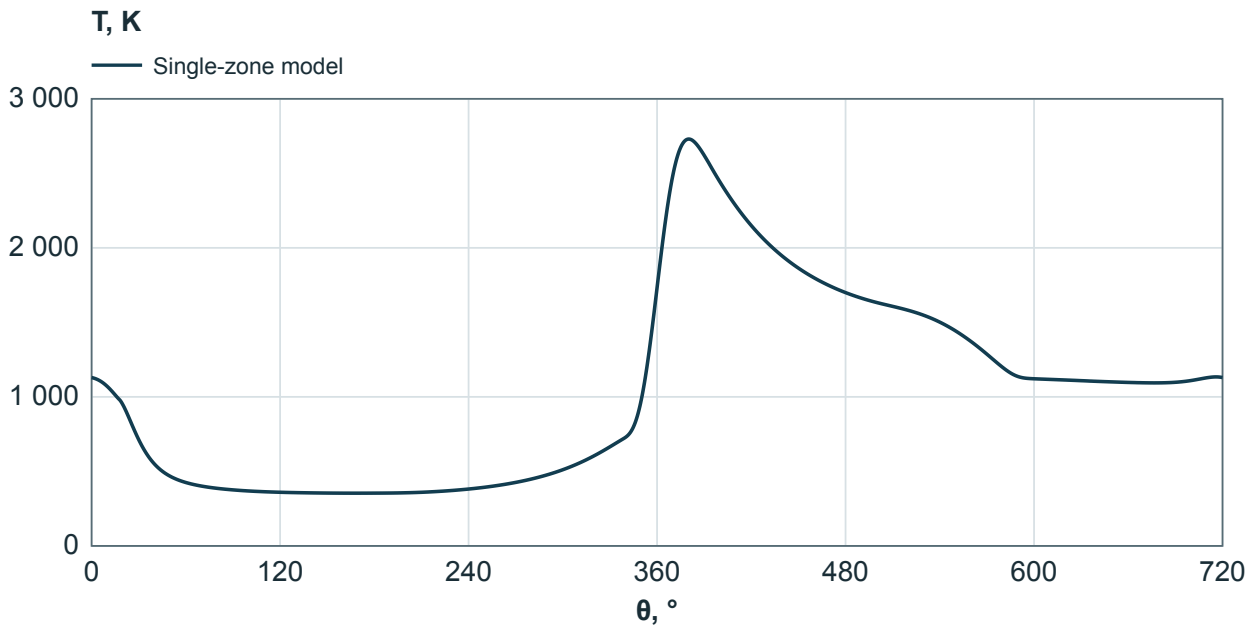


Figure 5.2 — Mean working-fluid temperature

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

6. Combustion and heat transfer

Heat release is a prescribed finite-duration function. Its start, duration and shape are separated. Spark timing is not automatically equated with combustion start; no separate ignition-delay model is included.

y = (θ−θ₀)/Δθ; x_b = [1−exp(−a·y^{m+1})]/[1−exp(−a)] for 0 < y < 1. (6.1) Before combustion x_b = 0; after its end x_b = 1. The Wiebe function comes from the literature model [4, 6]; the denominator normalisation in this project is an explicit adaptation to avoid counting η_c twice.

dQ_c/dθ = η_c m_f Hᵤ dx_b/dθ. (6.2) Shape coefficients, duration and start are assumptions. They were not obtained from a Toyota indicator diagram and do not establish optimal ignition timing.

Total energy before incomplete combustion is applied.

$$Q_f = m_f \cdot H_u \qquad (6.3)$$

$= (2{,}513 \cdot 10^{-5}\ \text{kg}) \cdot (4{,}400 \cdot 10^{7}\ \text{J/kg})$

= 1 105,6 J

The heat-release integral must match this quantity within quadrature error.

$$Q_c = \eta_c \cdot Q_f \qquad (6.4)$$

$= (0{,}98\ ) \cdot (1\ 105{,}6\ \text{J})$

= 1 083,48 J

h_w = 130·p_bar^0,8·T_K^(−0,4)·V_m³^(−0,06)·(c_m + 1,4)^0,8; dQ_w/dθ = h_w A_w(T−T_w)/(6n_rpm). (6.5) Hohenberg correlation is used in the units of equation (49) [6]. A_w = 2A_p + πD x is the area of two flat ends and the exposed wall. Valve surfaces, piston recesses and different wall temperatures are not represented separately.

p_{max} — Maximum calculated pressure: **6,647·10⁶ Pa**. Numerical snapshot

T_{max} — Maximum mean gas temperature: **2 730,03 K**. Numerical snapshot

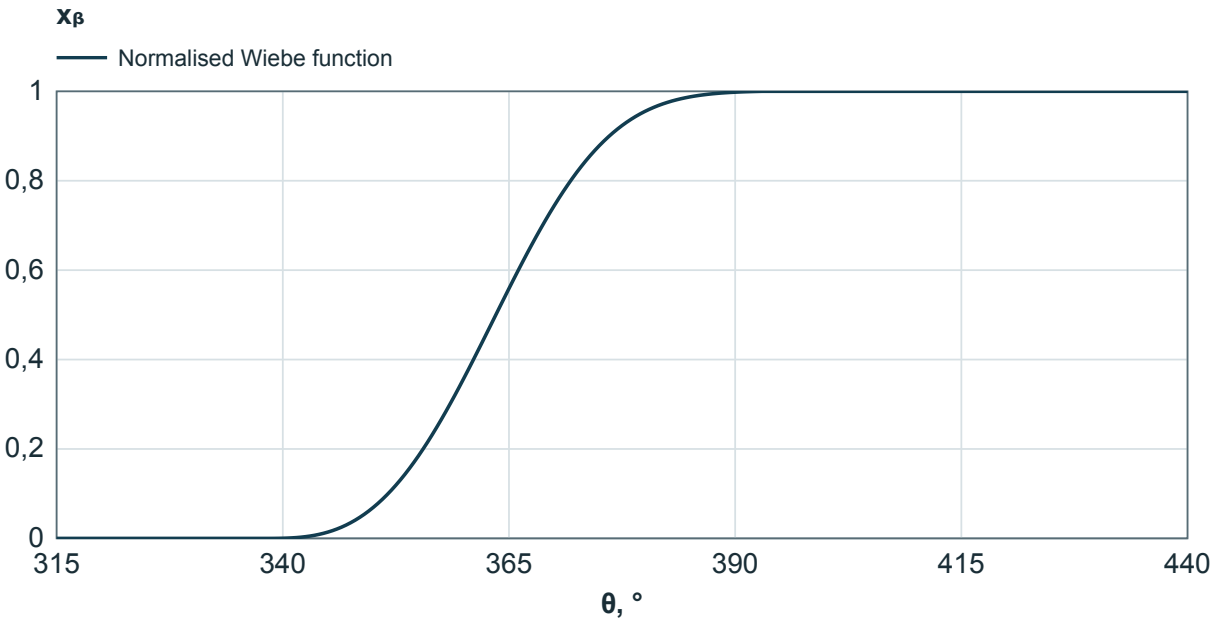


Figure 6.1 — Burned fraction of the model mixture

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

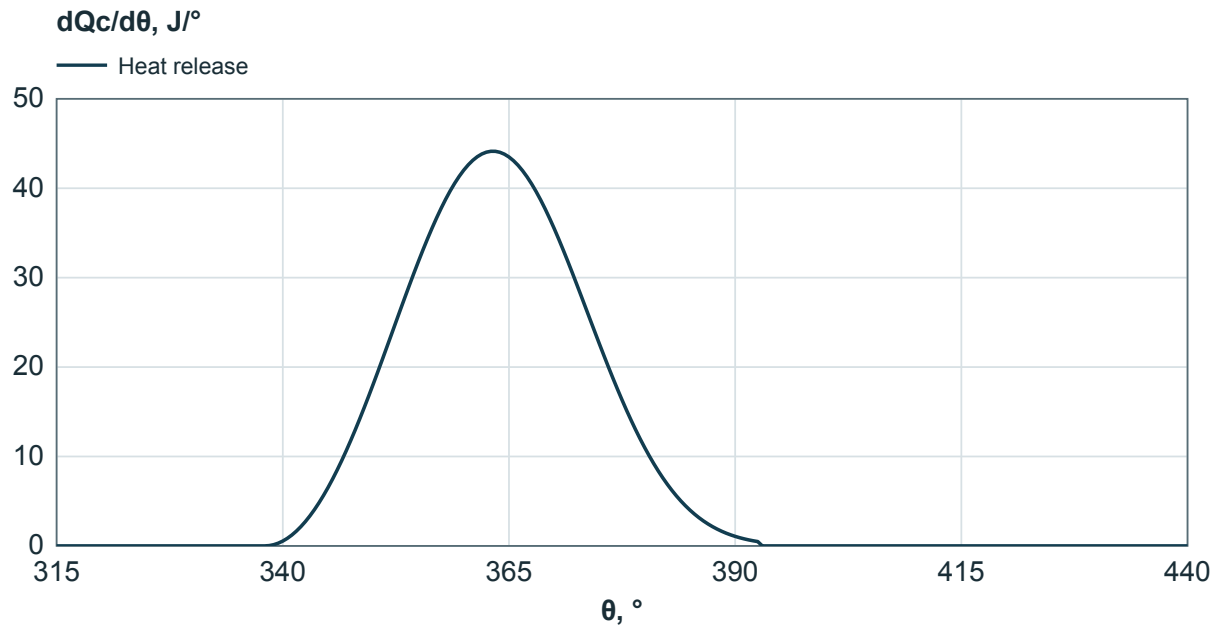


Figure 6.2 — Heat-release rate

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

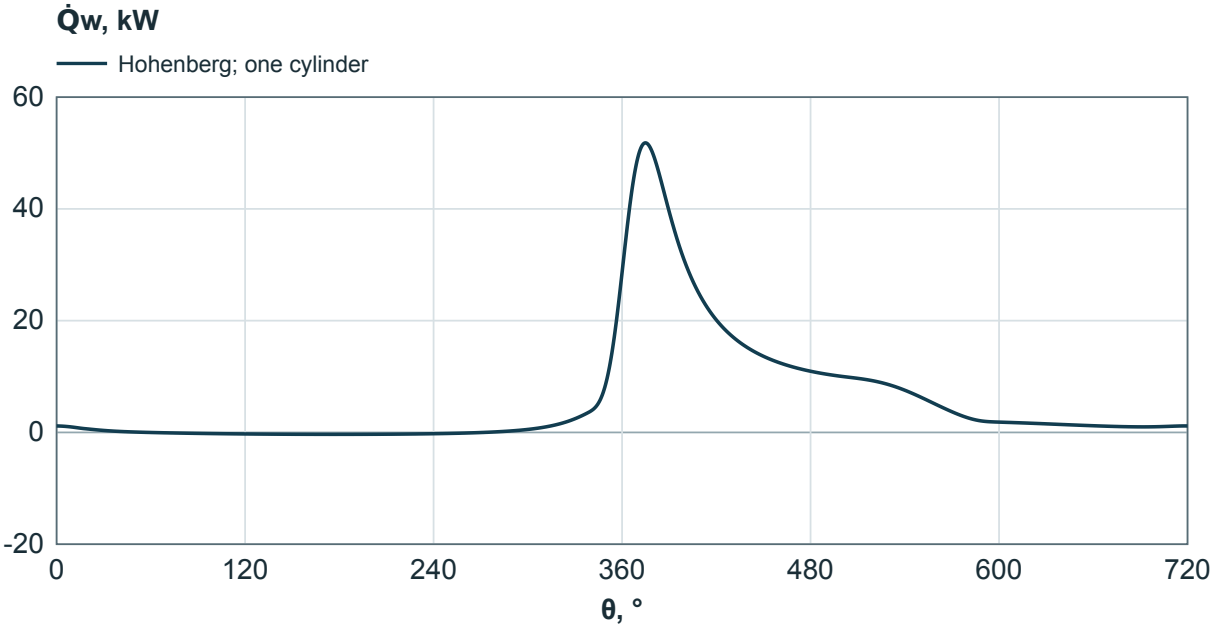


Figure 6.3 — Wall heat flow

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

7. Expansion, exhaust and indicator diagram

Expansion continues the same energy balance. After exhaust opening, blowdown and subsequent gas displacement are calculated, rather than imposing a vertical pressure drop at an artificial point.

p_b — Pressure at exhaust opening: **484 335 Pa**. Numerical snapshot

T_b — Temperature at exhaust opening: **1 632,75 K**. Numerical snapshot

W_i — Single-cylinder work integral over 720°: **458,067 J**. Numerical snapshot

W_p — Piston gas-exchange work over 0–180° and 540–720°: **-11,1126 J**. Numerical snapshot

W_i = ∮p dV. (7.1) W_i already includes gas exchange. Pumping losses must not be subtracted again. W_p separately shows piston gas-exchange work; late intake closure and early exhaust mean this is not an arbitrary alternative gross/net IMEP split.

The p(V) indicator diagram and expanded p(θ), T(θ) curves use the same arrays as integration. The low-pressure loop has its own scale. Maximum p and T may occur at different angles.

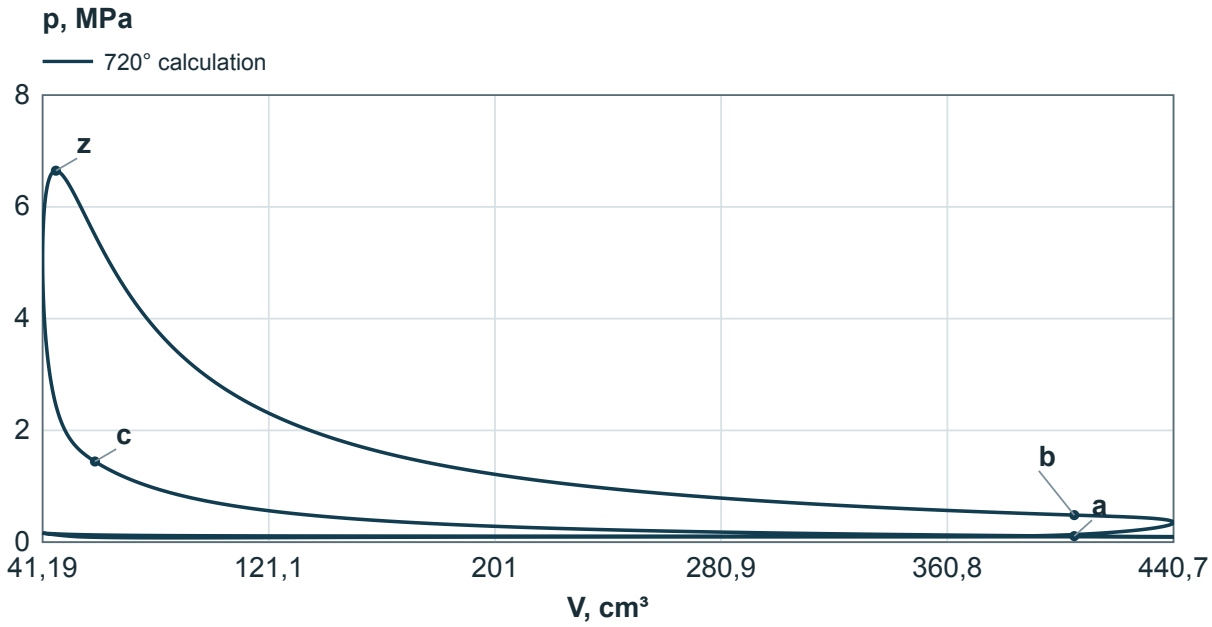


Figure 7.1 — Full-cycle indicator diagram

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

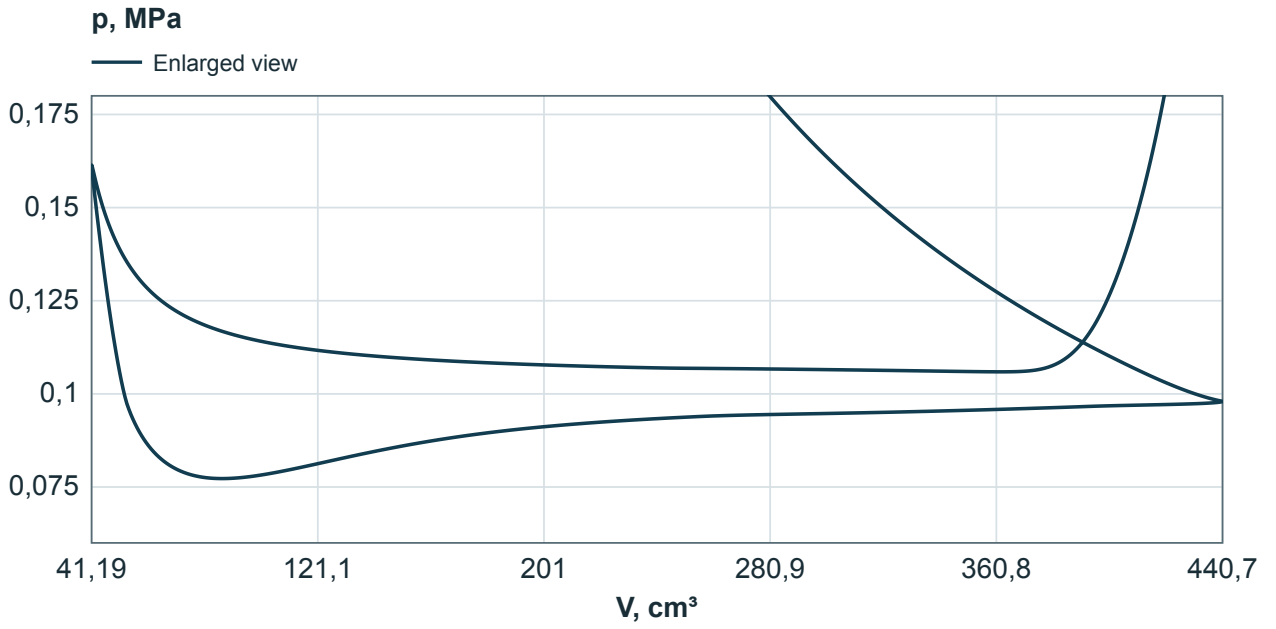


Figure 7.2 — Low-pressure p–V region

Model operating point: n = 4 400 min⁻¹; α = 1; рвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

8. Indicated and brake performance

Full-cycle indicated work, mechanical losses and useful shaft power are distinguished. Cooling and lubrication pump powers are not subtracted again: their detailed maps are unknown in the initial variant.

Integral metric from the calculated indicator diagram.

$$p_i = W_i / V_h \quad (8.1)$$

= (458,067 J) / (3,995·10⁻⁴ m³)

= 1,147·10⁶ Pa

A four-stroke engine completes one cycle every two revolutions.

$$P_i = W_i \cdot z \cdot n / 2 \quad (8.2)$$

= (458,067 J) · (4) · (73,3333 s⁻¹) / 2

= 67 183,1 W

A_m — Constant mechanical-loss term A: **40 000 Pa**. Adopted assumption

B_m — Mechanical-loss coefficient at p_max: **0,005** . Adopted assumption

C_m — Mechanical-loss coefficient at mean piston speed: **4 000 Pa·s/m**. Adopted assumption

D_m — Quadratic mechanical-loss coefficient: **100 Pa·s²/m²**. Adopted assumption

Adopted loss model; all four coefficients are assumptions, not test-bench results.

$$p_m = A_m + B_m \cdot p_{max} + C_m \cdot c_m + D_m \cdot c_m^2 \quad (8.3)$$

= (40 000 Pa) + (0,005) · (6,647·10⁶ Pa) + (4 000 Pa·s/m) · (11,5133 m/s) + (100 Pa·s²/m²) · (11,5133 m/s)²

= 132 544 Pa

p_m does not count gas-exchange pumping work a second time.

The same working-cycle frequency is used.

$$P_m = p_m \cdot V_h \cdot z \cdot n / 2 \quad (8.4)$$

= (132 544 Pa) · (3,995·10⁻⁴ m³) · (4) · (73,3333 s⁻¹) / 2

= 7 766,83 W

Useful power after adopted mechanical losses.

$$P_e = P_i - P_m \quad (8.5)$$

= (67 183,1 W) - (7 766,83 W)

= 59 416,3 W

Independent check of the power/torque relationship.

$$M_e = P_e / \omega \quad (8.6)$$

= (59 416,3 W) / (460,767 rad/s)

= 128,951 N·m

Fuel per cycle is converted into flow per second.

$$G_f = m_f \cdot z \cdot n / 2 \quad (8.7)$$

= (2,513·10⁻⁵ kg) · (4) · (73,3333 s⁻¹) / 2

= 0,00368532 kg/s

Initial heat-balance quantity.

$$P_f = G_f \cdot H_u \quad (8.8)$$

= (0,00368532 kg/s) · (4,400·10⁷ J/kg)

= 162 154 W

Not measured efficiency of a production engine.

$$\eta_e = P_e / P_f \quad (8.9)$$

= (59 416,3 W) / (162 154 W)

= 0,366419

To express the result in g/(kWh), multiply by 3,6·10⁹.

$$g_e = G_f / P_e \quad (8.10)$$

= (0,00368532 kg/s) / (59 416,3 W)

= 6,203·10⁻⁸ kg/J

Energy balance: P_f = P_e + P_m + Q̇_w + (Ḣ_out−Ḣ_in) + P_неповн + dU/dt. (8.11) The residual is calculated from all independently integrated components. It is not called “unaccounted losses” to force the total to 100 %. Q̇_w is heat through the cylinder walls, not the radiator's total thermal duty.

Table 8.1 — Heat-balance powers

Component	kW	% from P_f
Brake work	59,4163	36,642
Mechanical losses	7,76683	4,7898
Heat transferred to cylinder walls	23,9331	14,759
Net gas-flow enthalpy	67,7898	41,806
Chemical incompleteness	3,24808	2,0031
Energy change per cycle × cycles/s	-2,390·10⁻⁶	-1,474·10⁻⁶

Residual: 5,821·10⁻¹¹ W. Specific fuel consumption: **223,29 g/kWh**; η_i = 0,41432, η_e = 0,36642, η_m = 0,88439.

9. Engine speed characteristic

An independent periodic-cycle calculation is shown on a 1500–6400 min⁻¹ grid. Intake-reservoir pressure, valve timing and lift, and adopted heat release remain constant. This is a model speed characteristic at the specified maximum filling, not a reproduced manufacturer ECU map.

The speed characteristic relates power, torque and specific fuel consumption to rotational speed [7]. From two rated points [1] cannot reconstruct a reliable complete BSFC curve. Manufacturer values therefore appear only as individual markers where the corresponding quantity is known. The rated torque of 160 N·m is not extrapolated to other speeds.

P_e(n), M_e(n), g_e(n), G_f(n) are recalculated at each operating point. (9.1) Substantial deviation from specifications indicates a need to refine gas exchange, working fluid, combustion phasing and mechanical losses. Arbitrarily fitting all coefficients simultaneously does not make the model identifiable.

Table 9.1 — Calculated-characteristic operating points

n, min⁻¹	P_e, kW	M_e, N·m	g_e, g/(kWh)	Gf, kg/h
1 500	21,18	134,8	217,9	4,615
2 000	28,36	135,4	216,9	6,153
2 500	35,34	135	217,2	7,674
3 000	42,07	133,9	218,1	9,177
3 500	48,53	132,4	219,6	10,66
4 000	54,7	130,6	221,5	12,12
4 400	59,42	129	223,3	13,27
5 000	66,09	126,2	226,4	14,96
5 500	71,28	123,8	229,3	16,35
6 000	76,1	121,1	232,6	17,7
6 400	79,69	118,9	235,5	18,77

Table 9.2 — Comparison with two rated points [1]

Quantity	Specification	Model	Deviation, %
M_e at 4400 min⁻¹, N·m	160	129	-19,41
P_e at 6400 min⁻¹, kW	97	79,69	-17,85

This compares descriptions of different completeness, rather than validating the model. Changing geometry means comparing a different design.

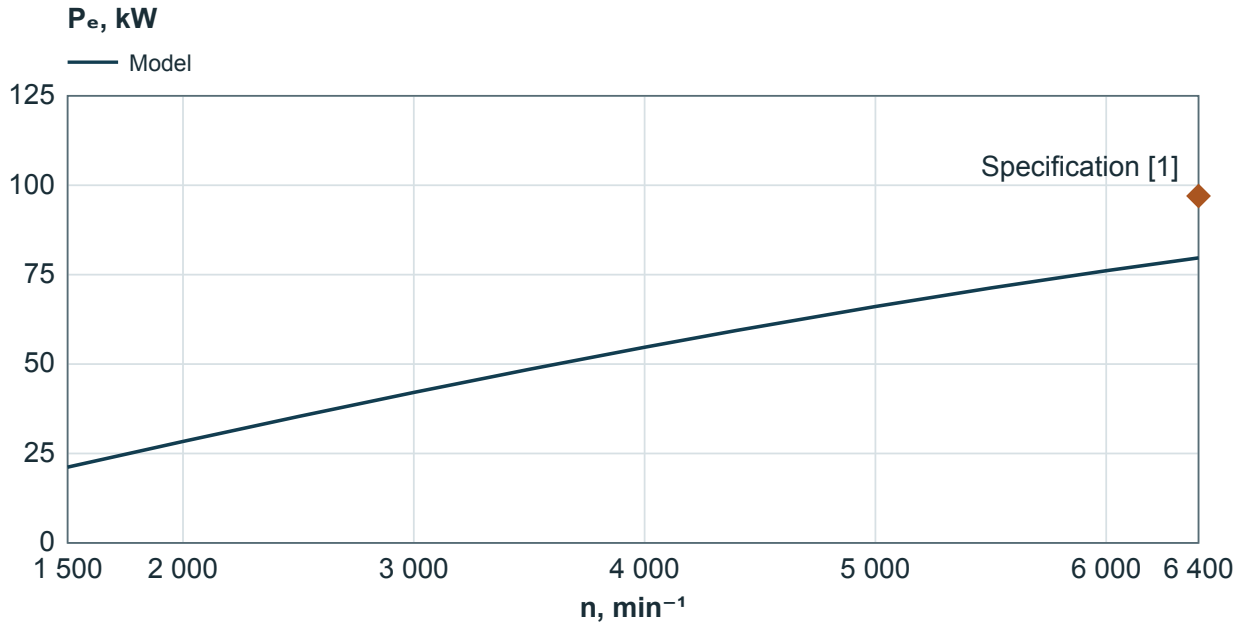


Figure 9.1 — Model power and rated point

Calculated series 1500–6400 min⁻¹; not a manufacturer map.

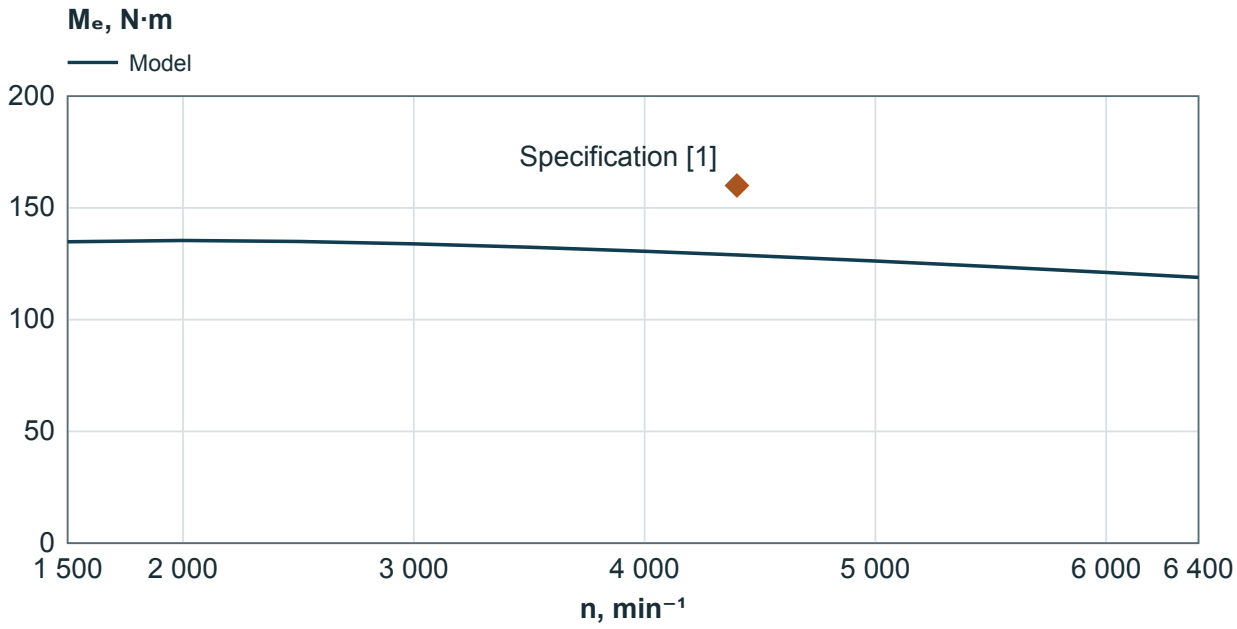


Figure 9.2 — Model torque and rated point

Calculated series 1500–6400 min⁻¹; not a manufacturer map.

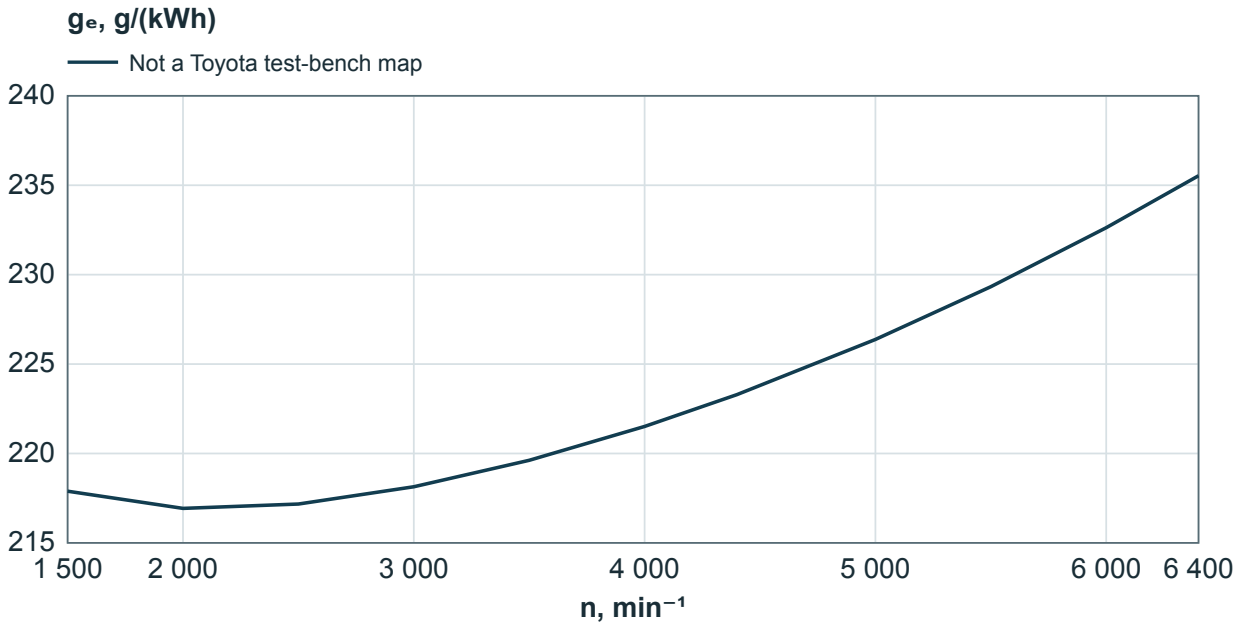


Figure 9.3 — Calculated specific fuel consumption

Calculated series 1500–6400 min⁻¹; not a manufacturer map.

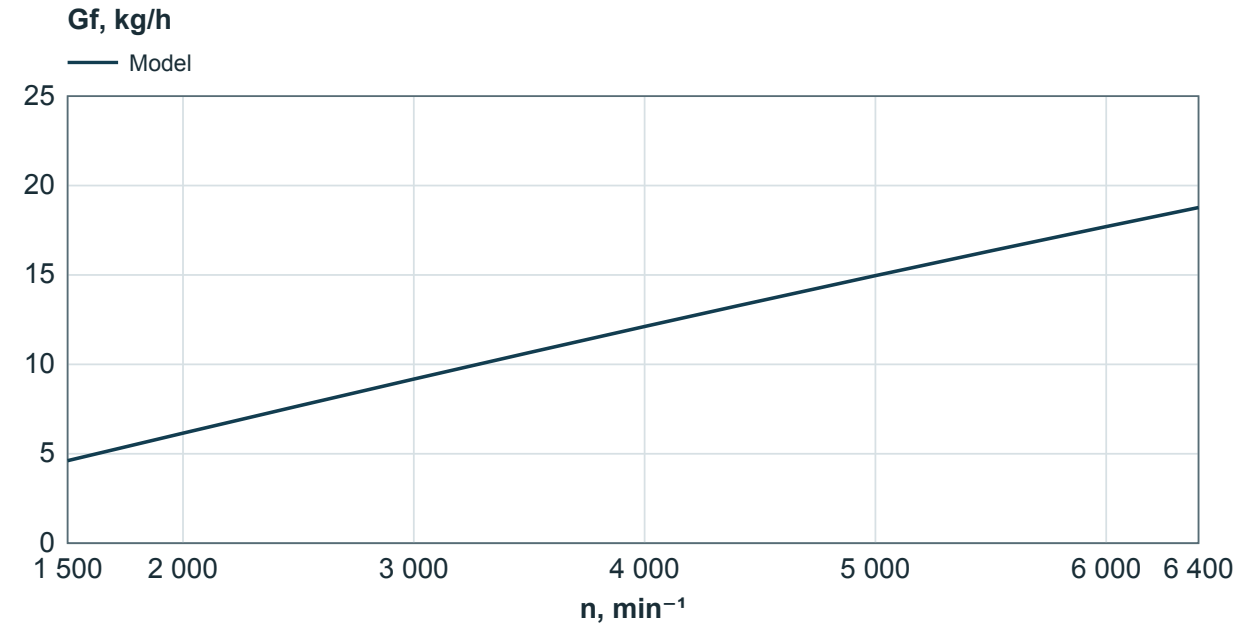


Figure 9.4 — Calculated hourly fuel consumption

Calculated series 1500–6400 min⁻¹; not a manufacturer map.

10. Slider-crank kinematics

Exact centred slider-crank geometry is used at constant angular speed. Pin offset, crankshaft offset, elastic deformation and torsional vibration are excluded. Connecting-rod length is an assumption.

$x(\theta) = r(1-\cos\theta) + L - \sqrt{(L^2 - r^2\sin^2\theta)}$. (10.1) x runs from TDC towards BDC. $V(\theta) = V_c + A_p\, x(\theta)$; $\sin\beta = (r/L)\sin\theta$. In the derivatives below, θ uses radians.

$dx/d\theta = r\sin\theta + r^2\sin\theta\cos\theta/\sqrt{(L^2 - r^2\sin^2\theta)}$; $v_p = \omega \cdot dx/d\theta$. (10.2)

$d^2x/d\theta^2 = r\cos\theta + r^2(\cos^2\theta - \sin^2\theta)/q + r^4\sin^2\theta\cos^2\theta/q^3$, where $q = \sqrt{(L^2 - r^2\sin^2\theta)}$; $a_p = \omega^2 \cdot d^2x/d\theta^2$. (10.3) Derivatives were checked independently with central differences. Kinematic diagrams repeat every 360°, gas-dynamic diagrams every 720°.

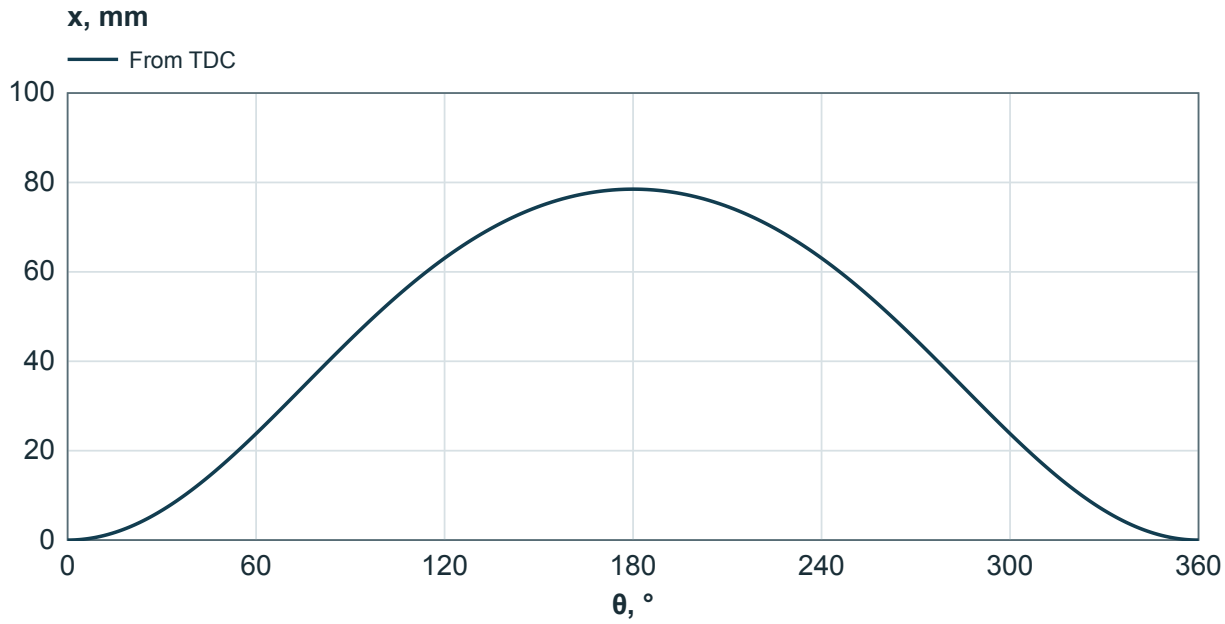


Figure 10.1 — Piston displacement

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ_0 = 338°; Δθ = 55°. Timing and masses are assumptions.

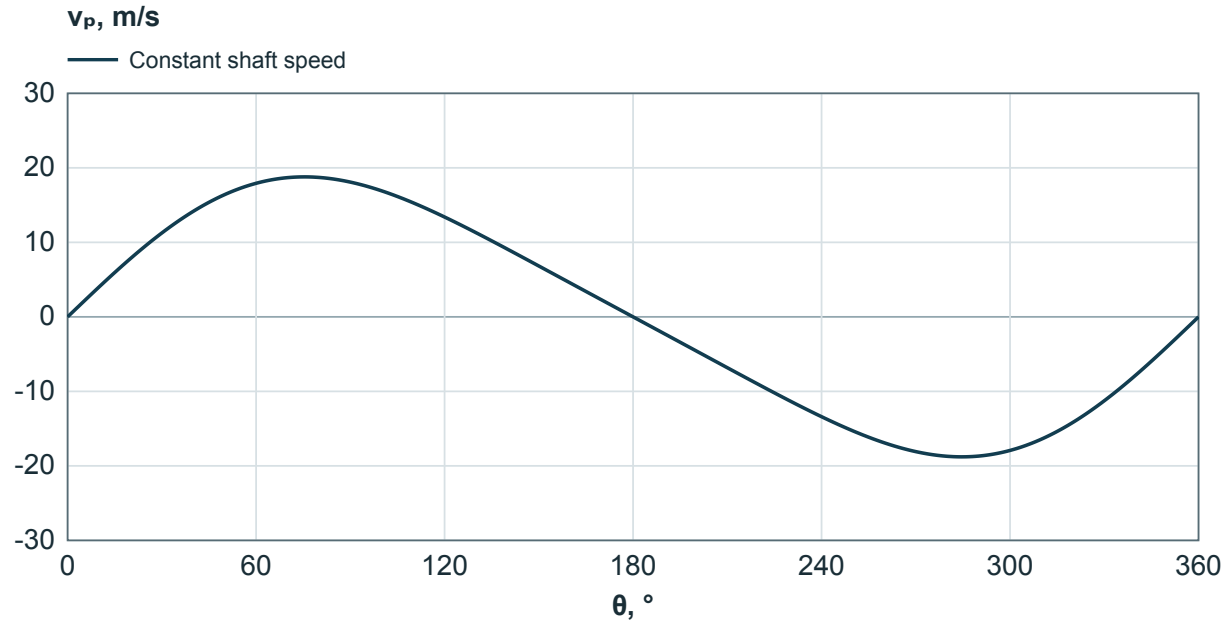


Figure 10.2 — Piston speed

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ_0 = 338°; Δθ = 55°. Timing and masses are assumptions.

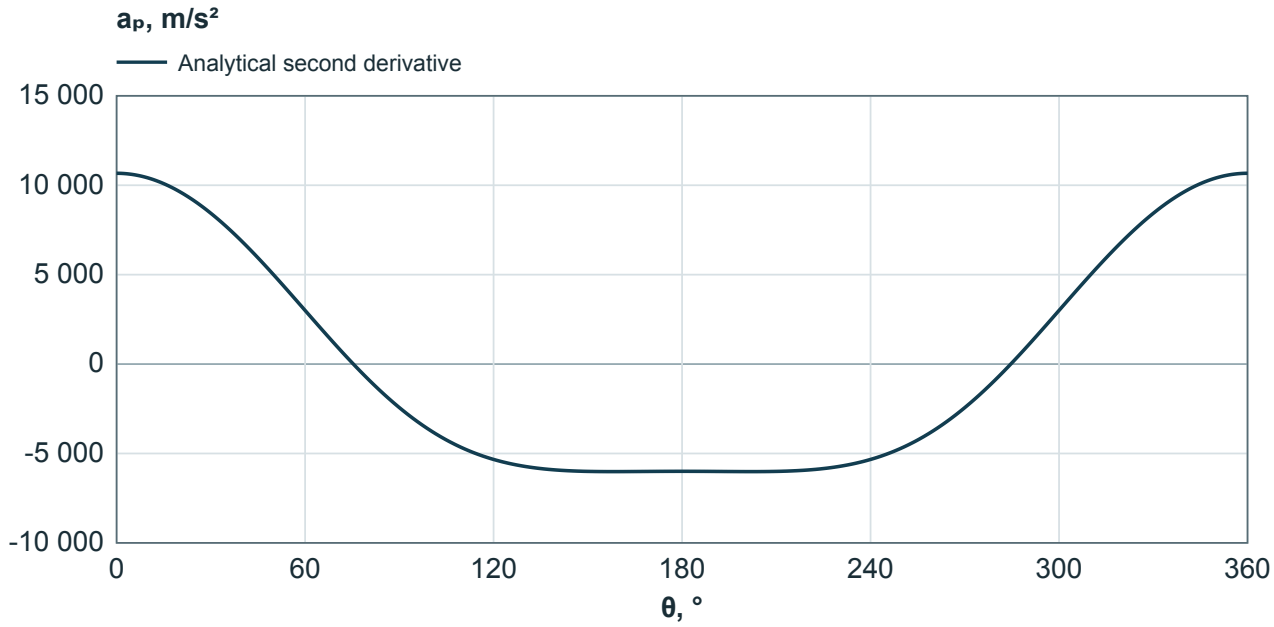


Figure 10.3 — Piston acceleration

Model operating point: n = 4 400 min⁻¹; α = 1; рвп = 98 kPa; θ_0 = 338°; Δθ = 55°. Timing and masses are assumptions.

11. Dynamic loads and torque

Forces follow current p(θ), the same geometry and the adopted equivalent mass. Slider-crank component functions and typical loading directions follow the teaching description [8], but numerical parameters from the old diesel example were not carried over.

m_j — Equivalent reciprocating mass per cylinder: **0,48 kg**. Adopted assumption

p_0 — Absolute crankcase pressure: **100 000 Pa**. Adopted assumption

Gas force acts on piston area; absolute pressure is not used without subtracting pressure below the piston.

$$F_{gmax} = (p_{max} - p_0) \cdot A_p \quad (11.1)$$

$= ((6{,}647 \cdot 10^6 \text{ Pa}) - (100\,000 \text{ Pa})) \cdot (0{,}00508958 \text{ m}^2)$

= 33 321,8 N

F_j(θ) = −m_j a_p(θ); F_Σ = F_g + F_j; F_ш = F_Σ/cosβ; N = F_Σ tanβ. (11.2) Positive axial force points with piston motion towards BDC. F_ш is signed connecting-rod compression; N is the side reaction under the adopted convention.

T = F_Σ·sin(θ+β)/cosβ; K = F_Σ·cos(θ+β)/cosβ; M_ц = rT. (11.3) T and K are tangential and radial components of the load transmitted by the connecting rod. They exclude centrifugal force from its rotating portion because that mass is unknown. K is not a completed bearing-load calculation.

M_Σ(θ) = Σ M_ц[θ−j·720°/z]. (11.4) Four cylinders are assumed evenly phased at 180°; cylinder numbering 1–3–4–2 is a working assumption. Single-cylinder torque includes inertia. Mean M_Σ is compared with P_i/ω, not M_e; the difference is the mean mechanical-loss torque.

Δ_E — Excess-energy range per cycle: **92,5787 J**. Numerical snapshot

δ — Permissible relative speed fluctuation: **0,02** . Project selection

Estimate using ΔE/(δω²) for small speed fluctuations.

$$J_{total} = \Delta_E / (\delta \cdot \omega^2) \quad (11.5)$$

$= (92{,}5787 \text{ J}) / ((0{,}02\ ) \cdot (460{,}767 \text{ rad/s})^2)$

= 0,0218031 kg·m²

This is the required total inertia, not extra flywheel inertia: the inertia of existing components is unknown.

Calculated load	Value
Maximum |Fш|, kN	28,39
Maximum |N|, kN	2,454
Maximum |Fj|, kN	5,121
Integral mean MΣ, N·m	145,8
P_i/ω, N·m	145,8
P_e/ω, N·m	129
Energy range, J	92,58

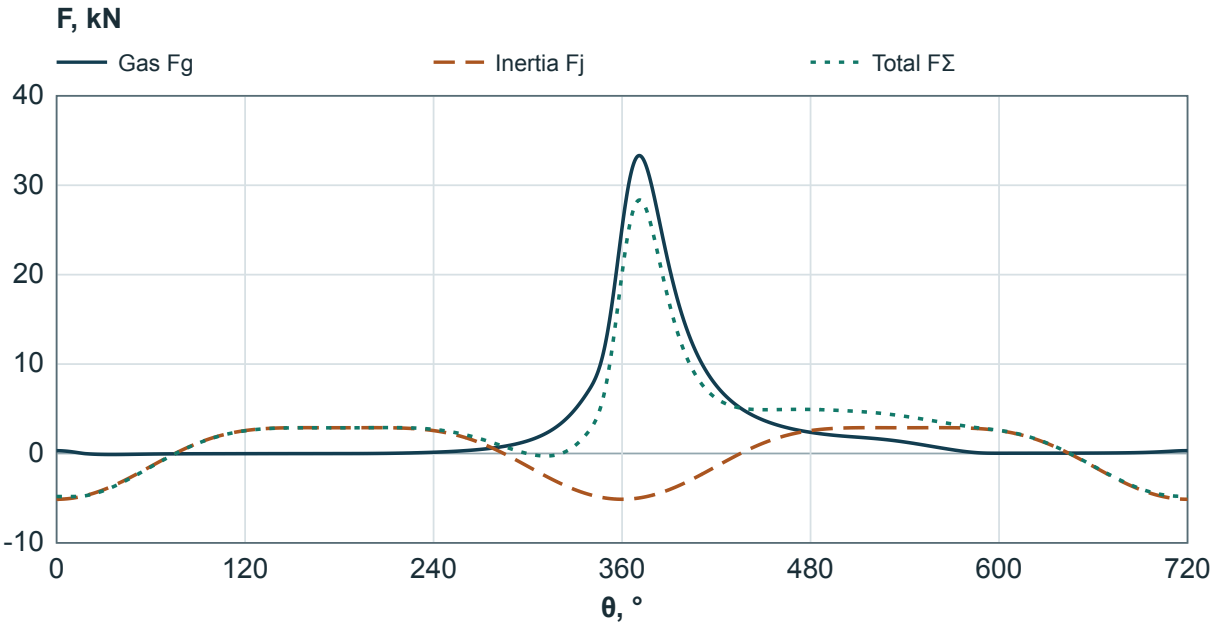


Figure 11.1 — Single-cylinder axial forces

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

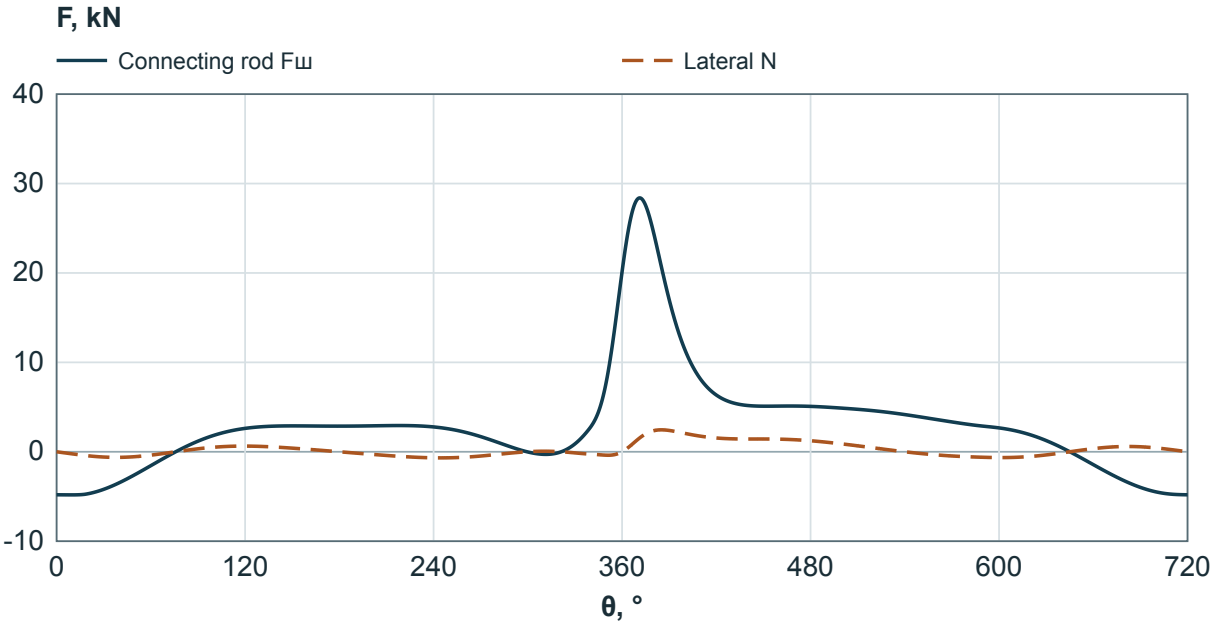


Figure 11.2 — Connecting-rod force and side reaction

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

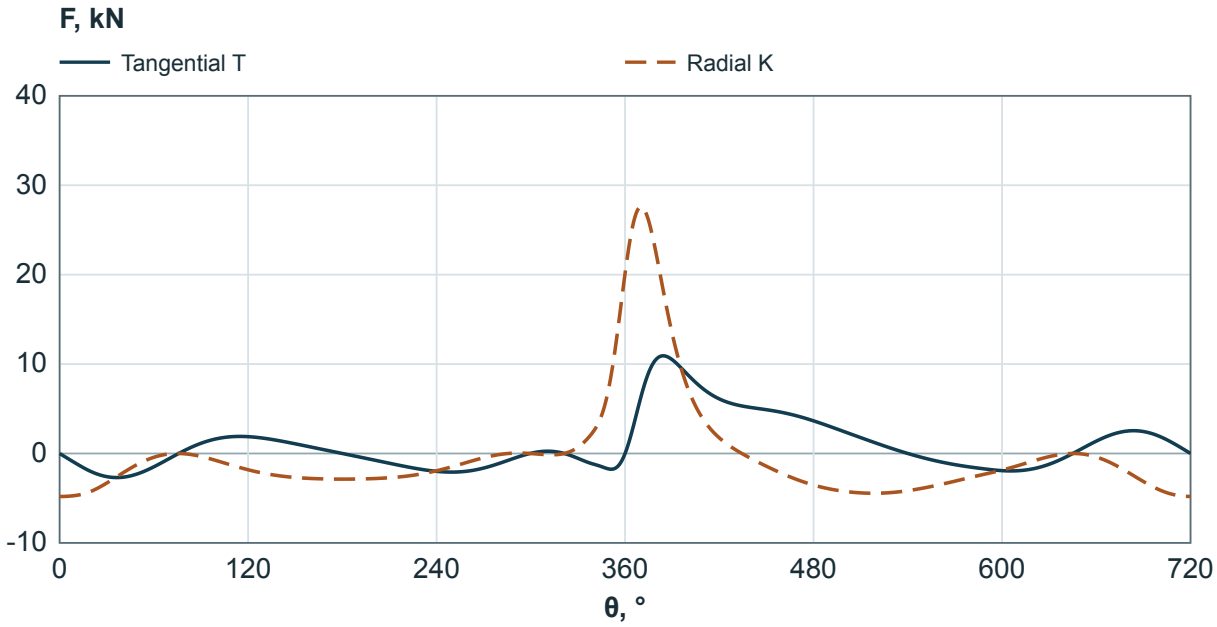


Figure 11.3 — Components of force transmitted by the connecting rod

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

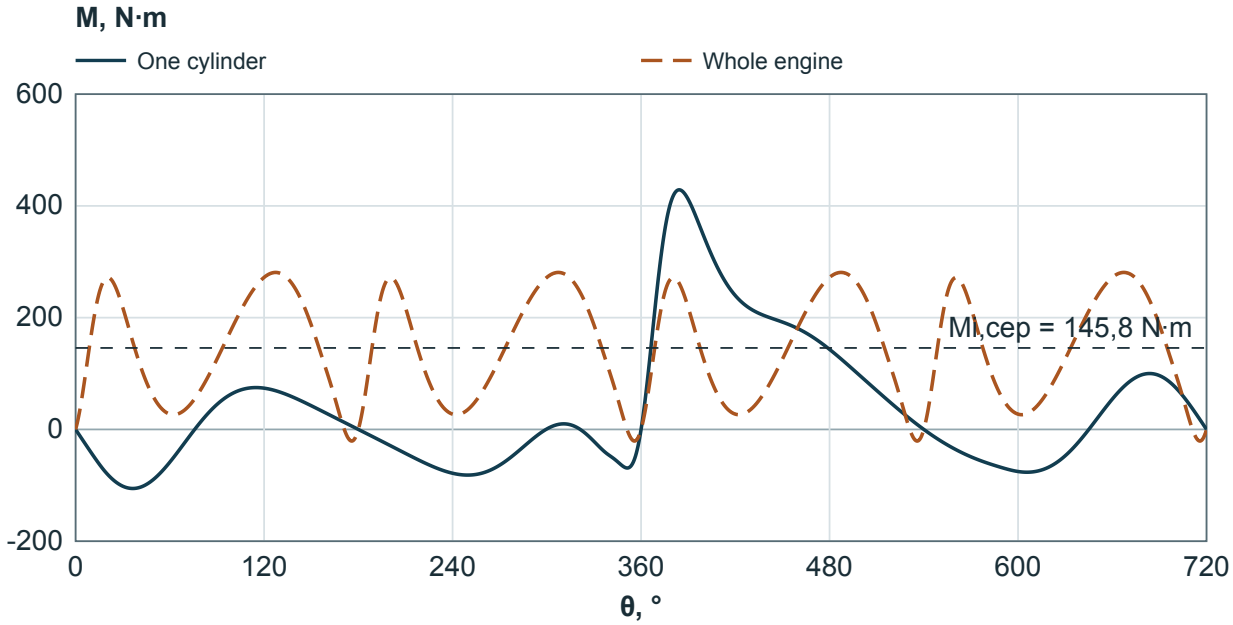


Figure 11.4 — Single-cylinder and whole-engine torque

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

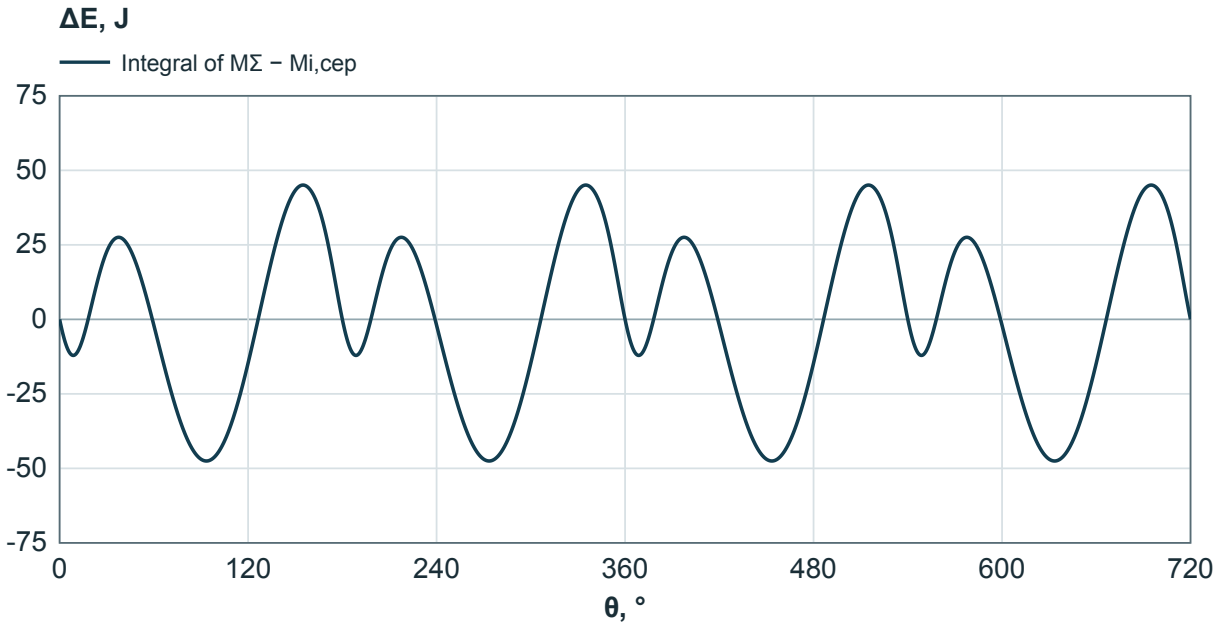


Figure 11.5 — Excess energy at constant speed

Model operating point: n = 4 400 min⁻¹; α = 1; pвп = 98 kPa; θ₀ = 338°; Δθ = 55°. Timing and masses are assumptions.

12. Preliminary auxiliary-system estimates

These estimates indicate the order of circulation flows from thermal power. They are not a finished selection of a pump, radiator, lubrication passages or valve mechanism. Hydraulic resistance, operating maps, hot-idle margins and temperature nonuniformity need separate inputs.

$Qdot_w$ — Heat transferred through cylinder walls per second: **23 933,1 W**. Numerical snapshot

χ — Fraction of mechanical losses entering the cooling circuit: **0,7** . Adopted assumption

ρ_c — Adopted coolant density: **1 050 kg/m³**. Adopted assumption

c_c — Adopted coolant specific heat capacity: **3 600 J/(kg·K)**. Adopted assumption

Δ_{Tc} — Permitted coolant temperature difference in the flow estimate: **10 K**. Adopted assumption

Explicit allocation of part of the mechanical losses to the cooling circuit; exhaust-port heat and other sources are not separately estimated.

$$Vdot_c = (Qdot_w + \chi \cdot P_m) / (\rho_c \cdot c_c \cdot \Delta_{Tc}) \quad (12.1)$$

= ((23 933,1 W) + (0,7) · (7 766,83 W)) / ((1 050 kg/m³) · (3 600 J/(kg·K)) · (10 K))

= 7,770·10⁻⁴ m³/s

ρ_o — Adopted oil density: **850 kg/m³**. Adopted assumption

c_o — Adopted oil specific heat capacity: **2 100 J/(kg·K)**. Adopted assumption

Δ_{To} — Adopted oil temperature difference: **15 K**. Adopted assumption

Thermal estimate using the adopted loss fraction only.

$$Vdot_o = (1 - \chi) \cdot P_m / (\rho_o \cdot c_o \cdot \Delta_{To}) \quad (12.2)$$

= (1 - (0,7)) · (7 766,83 W) / ((850 kg/m³) · (2 100 J/(kg·K)) · (15 K))

= 8,702·10⁻⁵ m³/s

Total oil-pump flow also depends on leakage, pressure, bearing lubrication and piston-cooling jets.

This version does not claim verified strength of the piston, pin, connecting rod, rod bolts or crankshaft. Calculation requires section dimensions, materials, heat treatment, limiting load cycles, actual masses and allowable stresses. Available maximum forces are loads for the next stage, not proof of a safe design.

In practical units: Estimated coolant circulation 46,62 L/min; oil 5,221 L/min. These are not rated Toyota pump flows.

13. Checks, reproduction and status

Software verification and physical validation are different. Numerical balances, convergence and constant invariants may pass even when the real-engine model is inaccurate.

Checks cover positive mass and temperature, periodic-cycle closure, mass and energy balance, monotonic x_b, independent ∮p dV quadrature, torque integral, kinematic derivatives, scalar-formula dimensions, absence of dependency cycles and compliance with the current engdoc-native-draft/0.1. Check results apply only to this project version.

Explanations are stored as small JSON fragments in manuscript/. Assembly produces one valid native document. Custom chart nodes are not invented for the native schema. Integrator arrays and the figure register go to the publication module, which generates HTML and PDF from the same data.

The structural validator is unchanged from Text_work dated 23.09.2026. Scalar ICE calculations use a separate restricted SI adapter, not the original mathjs-safe. Integrator input snapshots are marked snapshot. Changing physical parameters requires full recalculation in ICE: Text_work does not run the 720° integrator itself. This project does not confirm complete installation or tests of the new Text_work dependency build.

A1 is a vector-figure appendix in a 3 × 2 layout. A4 and A1 use one register and identical arrays. The layout is an adopted working variant: the exact sample in Doc2.rar was not read. Text and PDF sources are stored separately; unread DJVU files are not marked verified.

Check current calculation	Obtained
Repeats to periodic state	4
Relative periodic residual	$7{,}762 \cdot 10^{-6}$
Mass residual, kg	$-3{,}253 \cdot 10^{-19}$
Energy residual, J/cycle	$-5{,}222 \cdot 10^{-13}$
Relative error of independent work quadrature	$1{,}684 \cdot 10^{-5}$
Relative work/torque discrepancy	$6{,}217 \cdot 10^{-10}$

Appendix A. Complete parameter register

All values of the evaluated state. Interface limits safeguard the numerical variant; they are not manufacturer tolerances.

Parameter	Value	Unit	Basis
Cylinder bore	0,0805	m	Manufacturer data [2]
Piston stroke	0,0785	m	Manufacturer data [2]
Geometric compression ratio	10,7	—	Manufacturer data [2]
Number of cylinders	4	—	Manufacturer data [2]
Connecting rod centre distance	0,14	m	Adopted assumption
Equivalent reciprocating mass per cylinder	0,48	kg	Adopted assumption
Crankshaft rotational speed	4 400	1/min	Project selection
Absolute intake-reservoir pressure	98 000	Pa	Adopted assumption
Fresh-mixture temperature in the intake reservoir	315	K	Adopted assumption
Absolute exhaust-reservoir pressure	105 000	Pa	Adopted assumption
Gas temperature during exhaust backflow	850	K	Adopted assumption
Absolute crankcase pressure	100 000	Pa	Adopted assumption
Effective combustion-chamber wall temperature	430	K	Adopted assumption
Intake closure relative to gas-exchange TDC	220	deg	Adopted assumption
Exhaust opening relative to gas-exchange TDC	500	deg	Adopted assumption
Assumed intake-valve diameter	0,032	m	Adopted assumption
Assumed exhaust-valve diameter	0,028	m	Adopted assumption
Assumed maximum intake-valve lift	0,009	m	Adopted assumption
Assumed maximum exhaust-valve lift	0,008	m	Adopted assumption
Valve flow coefficient	0,7	—	Adopted assumption
Excess-air ratio	1	—	Project selection
Stoichiometric air-to-fuel mass ratio	14,956	—	Literature example [3]
Lower heating value of model gasoline	$4{,}400 \cdot 10^{7}$	J/kg	Literature example [3]
Model heat-release start	338	deg	Adopted assumption
Model heat-release duration	55	deg	Adopted assumption
Wiebe-function parameter a	6,908	—	Adopted assumption [4]

Parameter	Value	Unit	Basis
Wiebe-function shape exponent m	2	—	Adopted assumption
Fraction of chemical energy converted to heat	0,98	—	Adopted assumption
Hohenberg correlation multiplier	1	—	Project selection
Constant mechanical-loss term A	40 000	Pa	Adopted assumption
Mechanical-loss coefficient at p_max	0,005	—	Adopted assumption
Mechanical-loss coefficient at mean piston speed	4 000	Pa*s/m	Adopted assumption
Quadratic mechanical-loss coefficient	100	Pa*s^2/m^2	Adopted assumption
Fraction of mechanical losses entering the cooling circuit	0,7	—	Adopted assumption
Adopted coolant density	1 050	kg/m^3	Adopted assumption
Adopted coolant specific heat capacity	3 600	J/(kg*K)	Adopted assumption
Permitted coolant temperature difference in the flow estimate	10	K	Adopted assumption
Adopted oil density	850	kg/m^3	Adopted assumption
Adopted oil specific heat capacity	2 100	J/(kg*K)	Adopted assumption
Adopted oil temperature difference	15	K	Adopted assumption
Permissible relative speed fluctuation	0,02	—	Project selection
Main angular integration step	0,5	deg	Project selection
Maximum cycle repeats	20	—	Project selection
Mass/energy periodic-convergence criterion	$1{,}000 \cdot 10^{-5}$	—	Project selection

Source list

[1] Toyota. The New Toyota Auris. 2 March 2015. Sections Revised powertrain line-up and 1.6-litre Valvematic.
Use and limits: Euro 6 for the Auris 2015 range; 97 kW / 6400 min^{-1}, 160 N·m / 4400 min^{-1}. No complete test-bench map is available. Accessed: 23.09.2026.

[2] Toyota. The New Toyota Auris. 2012. Technical specifications table: 1ZR-FAE, 1.6 Valvematic.
Use and limits: 80,5 × 78,5 mm; ε = 10,7; 4 cylinders, 1598 cm^3. This supplies geometry from a previous version in the same family, not specifications of the particular Euro 6 engine. Accessed: 23.09.2026.

[3] Example gasoline-engine thermal-cycle calculation. User-supplied PDF; author and year could not be established from the file. 10 pp.; items 1–19, particularly pp. 1, 4, 7–10.
Use and limits: calculation sequence; model gasoline l_0 = 14,956, H_u = 44 MJ/kg. Values do not certify modern E5/E10. Accessed: 23.09.2026.

[4] Van Basshuysen R., Schäfer F. (eds.). Internal Combustion Engine Handbook. Supplied excerpt. Printed pp. 452–455 and 471–472; original PDF: pp. 35–38, 54–55.
Use and limits: Wiebe function and the need to adapt it. Diesel-chapter coefficients are not adopted as Toyota calibration. Accessed: 23.09.2026.

[5] NASA Glenn Research Center. Mass Flow Rate Equations; Isentropic Flow Equations. Maximum flow at M = 1; isentropic relations.
Use and limits: flow limitation at M = 1. The Ψ formula is derived from continuity and isentropic relations; ambiguous MathJax notation for intermediate powers is not copied. Accessed: 23.09.2026.

[6] Chandrakar C. K., Varma P. S., Mittal M. Thermodynamics based modeling of various in-cylinder processes and spark ignition engine cycle with a focus on sub-models assessment. Scientific Reports. 2025. DOI: 10.1038/s41598-025-15819-7. Equations (49) Hohenberg; (69) Wiebe.
Use and limits: heat-transfer correlation form and units. The article's two-zone model is not reproduced, and its experimental validation is not claimed. Accessed: 23.09.2026.

[7] Sliusarov O. S. Methodical Guide for the Course Project “Vehicle Operational Properties”. Zaporizhzhia: ZNTU, 2017. 50 pp. Section 1.1, s. 5–7.
Use and limits: definition of the engine speed characteristic and the torque/power relationship. This is not a complete thermodynamic-cycle method. Accessed: 23.09.2026.

[8] Yanovych Yu. V. Construction of Automotive and Tractor Engines. Methodical Guide for Laboratory and Practical Classes. Supplied PDF: 31 pp.; laboratory exercises 1–8.
Use and limits: list of mechanisms, systems and terms; does not prescribe an A1 sheet layout. Accessed: 23.09.2026.